

Shari's Week 0 Day 3 Visualization Report

Basic Data Visualisation with Matplotlib

Mercer General Hospital | Clinical AI & Innovation Unit

Overview

For the Week 0 Day 3 task, the focus was on creating clinically meaningful plots from the cleaned Mercer General Emergency Department triage dataset using matplotlib. During the tutorial, we were introduced to different plot types, including bar plots, histograms, scatter plots, and box plots. The main lesson was that each plot should answer a specific clinical or data question, rather than simply display values.

For the individual student task, I created my own visualisations using the cleaned triage dataset. The required deliverables were a notebook containing at least one histogram and one scatter plot, with proper titles, labelled axes, clinical reference lines where appropriate, and saved plot images.

My two main challenge plots were:

1. A scatter plot showing the relationship between pulse and mean arterial pressure.
2. A histogram showing the distribution of systolic blood pressure values.

Day 3 Deliverables

The Day 3 submission required:

- One histogram with clinical reference lines labelled
- One scatter plot with labelled axes and a title connected to a clinical question
- Saved plot image outputs
- A notebook pushed to GitHub
- A plot image posted to Discord

My notebook met these requirements by including a scatter plot of pulse and MAP, a histogram of SBP, clinical reference lines and shaded zones, titles, axis labels, legends, saved image files, and short interpretations of each plot.

1. Matplotlib Setup

The notebook used matplotlib for plotting. The main plotting library was imported as:

```
import matplotlib.pyplot as plt
```

This allowed me to use functions such as `plt.subplots()`, `ax.scatter()`, `ax.hist()`, `ax.axvline()`, `ax.axhline()`, `plt.savefig()`, and `plt.show()`.

A standard plotting structure used in the notebook was:

```
fig, ax = plt.subplots(figsize=(8, 5))
# plotting code
ax.set_title("Plot Title")
ax.set_xlabel("X-axis label")
ax.set_ylabel("Y-axis label")
plt.tight_layout()
plt.savefig("plot_name.png", dpi=100, bbox_inches="tight")
plt.show()
```

This structure helped ensure that each plot was readable, labelled, and saved properly for submission.

2. Scatter Plot: Pulse and MAP

Clinical Question

The clinical question for this plot was:

Do patients with lower MAP tend to have higher pulse rates, which may suggest possible physiological stress or compensation?

This question was chosen because MAP and pulse are both important vital signs in emergency care. MAP gives an indication of blood perfusion, while pulse reflects heart rate. A patient with low MAP and high pulse may need closer review because this combination can suggest physiological stress, poor perfusion, dehydration, shock, pain, fever, or other acute conditions.

Plot Type Used

A scatter plot was used because both variables are numeric. Each point on the plot represents one patient, with pulse plotted on the x-axis and MAP plotted on the y-axis.

The main code used was:

```
fig, ax = plt.subplots(figsize=(7, 5))
ax.scatter(df["pulse"], df["MAP"], alpha=0.5, color="blue",
           edgecolors="black")
```

The $\alpha=0.5$ setting made the points slightly transparent, which helped with visibility where points overlapped.

Clinical Reference Lines and Annotation

Two clinical reference lines were added:

- MAP = 65 mmHg
- Pulse = 100 bpm

The code used was:

```
ax.axhline(y=65, color="red", linestyle=":", linewidth=1,
           label="MAP = 65 mmHg clinical review threshold")
ax.axvline(x=100, color="green", linestyle=":", linewidth=1,
           label="Pulse = 100 bpm typical adult range")
```

The horizontal line marked the MAP threshold, while the vertical line marked the pulse threshold. Together, these lines created a review zone for patients with both low MAP and high pulse.

An annotation was also added to label the review zone:

```
ax.text(128, 43, "Review zone:\nLow MAP + high pulse", fontsize=10,
       bbox=dict(facecolor="white", alpha=0.7, edgecolor="none"))
```

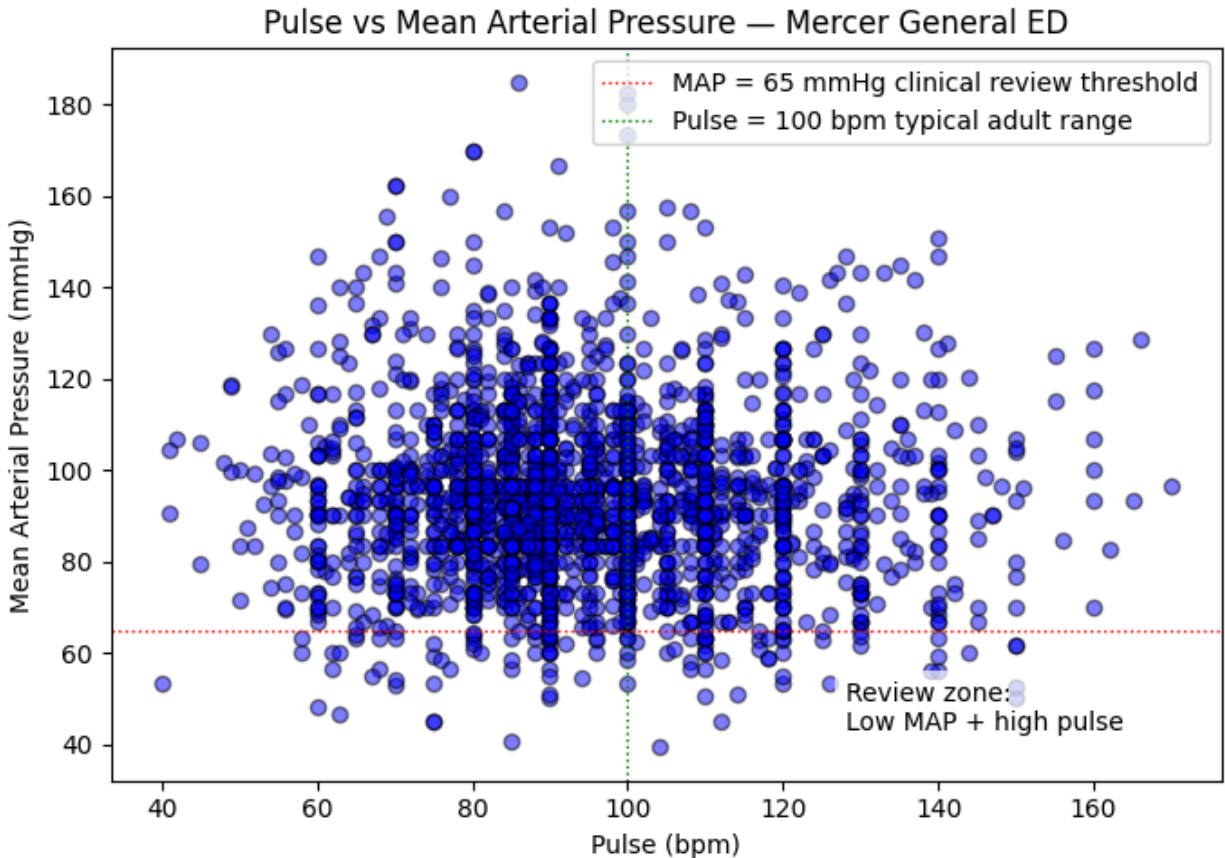
Interpretation

The scatter plot showed a broad spread of pulse and MAP values. The reference lines made it easier to identify patients who may require closer review, especially those with MAP below 65 mmHg and pulse above 100 bpm.

However, these values were not treated automatically as errors. In an emergency department dataset, abnormal values may represent real clinical presentations. For example, a patient may have an elevated pulse due to pain, fever, anxiety, dehydration, or shock. Therefore, the plot was used to flag possible clinical concern, not to remove data.

The plot was saved using:

```
plt.savefig("pulse_vs_map.png", dpi=100, bbox_inches="tight")
```



3. Histogram: Distribution of SBP Values

Clinical Question

The clinical question for this plot was:

What is the distribution of systolic blood pressure values among patients in the Mercer General ED triage dataset?

This question was chosen because systolic blood pressure is a key vital sign used in emergency assessment. It can help identify patients with hypotension, severe hypertension, or possible cardiovascular instability.

Plot Type Used

A histogram was used because SBP is a numeric variable. A histogram is appropriate when the goal is to understand the distribution of one clinical measurement across the dataset.

The main code used was:

```
fig, ax = plt.subplots(figsize=(7, 5))
ax.hist(df["SBP"].dropna(), bins=25, color="pink",
        edgecolor="black", alpha=0.8)
```

The `.dropna()` function removed missing values before plotting, while `bins=25` divided the SBP values into intervals to show the distribution clearly.

Clinical Reference Zones

To make the histogram clinically meaningful, shaded reference zones were added:

- SBP below 90 mmHg
- SBP above 180 mmHg

The code used was:

```
ax.axvspan(0, 90, color="green", alpha=0.08, label="Low SBP zone (<90 mmHg)")
ax.axvspan(180, 260, color="blue", alpha=0.08, label="High SBP zone (>180 mmHg)")
```

Reference lines were also added at the thresholds:

```
ax.axvline(90, color="green", linestyle=":", linewidth=1.5)
ax.axvline(180, color="blue", linestyle=":", linewidth=1.5)
```

SBP below 90 mmHg may suggest hypotension or shock risk, while SBP above 180 mmHg may suggest severe hypertension and may require clinical review.

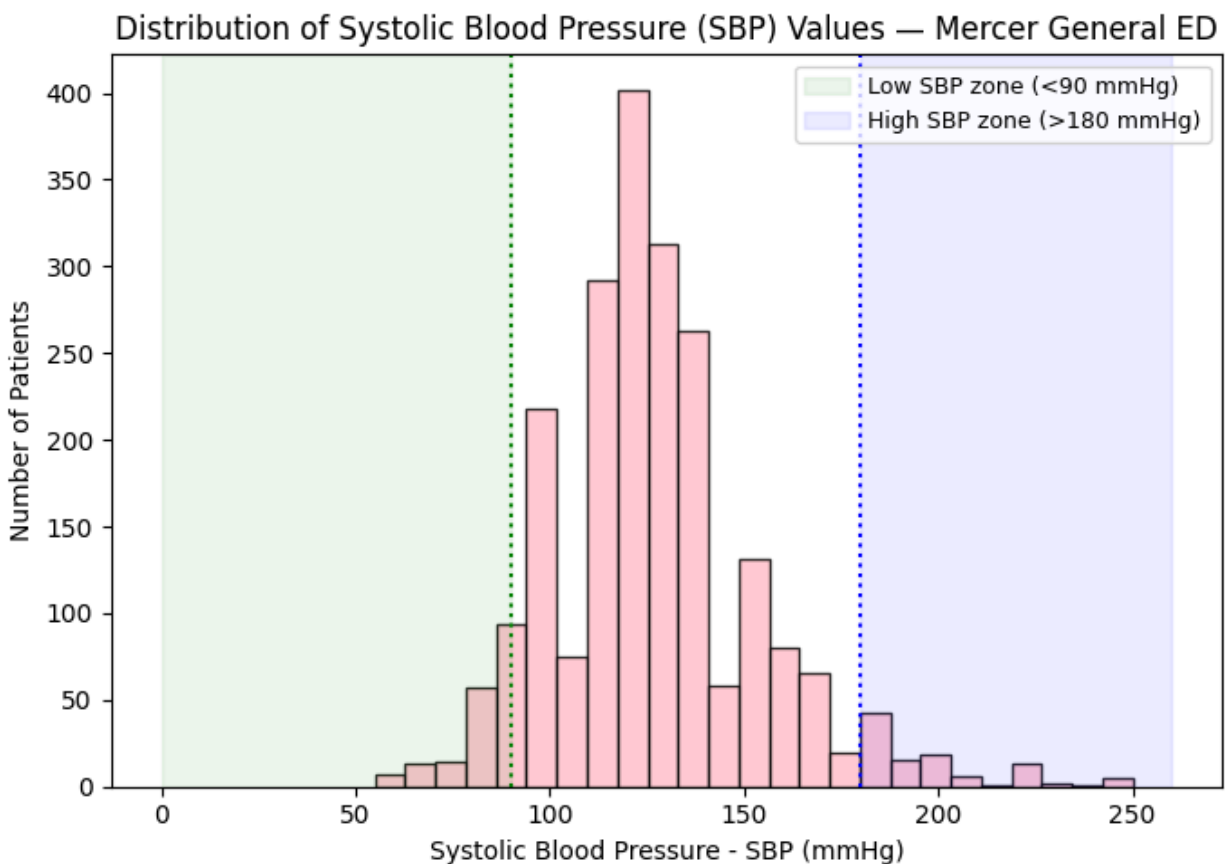
Interpretation

The SBP histogram showed that most patients had systolic blood pressure values clustered around the middle of the distribution, with fewer patients at very low or very high values.

The shaded zones helped identify patients whose SBP values may need further clinical review. However, these values should not automatically be treated as errors. In an emergency department setting, abnormal blood pressure values can be clinically real and may reflect the patient's presenting condition.

The plot was saved using:

```
plt.savefig("sbp_distribution_histogram.png", dpi=100,  
bbox_inches="tight")
```



4. Key Considerations

Several considerations guided the completion of the individual task.

First, the plots were based on the cleaned triage dataset. This was important because using cleaned data reduces the chance of visualizing values that had already been identified as invalid or inconsistent.

Second, the plots were selected based on clinical relevance. I selected MAP, pulse, and SBP because these are important vital signs in emergency assessment and can help identify patients who may need further clinical review.

Third, clinical reference lines and zones were added to improve interpretation. These included MAP = 65 mmHg, pulse = 100 bpm, SBP = 90 mmHg, and SBP = 180 mmHg.

Finally, abnormal values were interpreted carefully. In clinical data, unusual values may represent real patients rather than errors. Therefore, the plots were used to highlight values for review rather than automatically remove them.

Conclusion

For the Day 3 visualisation task, I created clinically meaningful plots using matplotlib and the cleaned Mercer General ED triage dataset. My individual task focused on producing a scatter plot and a histogram that answered specific clinical questions.

The scatter plot explored the relationship between pulse and MAP and highlighted a possible review zone for patients with low MAP and high pulse. The histogram showed the distribution of SBP values and used shaded reference zones to identify potentially low or high blood pressure values.

Overall, this task helped me understand how data visualisation can support clinical interpretation. It showed that plots should not only be clear and labelled, but should also help identify patterns, possible risks, and values that may require clinical review.